

A Proof-of-concept of Smart Hangar for Composite Aircraft

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ABSTRACT

From the past decades, the usage of composite materials in aircraft structures has been increased. However, with increasing usage of composite materials in aircraft structures, damage cases of the composite structures also are occurring such as barely visible impact damage (BVID), delamination, debonding, and fiber-matrix breakage. To enhance the efficiency of the aircraft maintenance performance, a new cutting-edge paradigm of structural health management – Smart Hangar for aerospace industry is proposed in this paper. The Smart Hangar is incorporated with an advanced ultrasonic propagation imaging (UPI) system to facilitate the full-scale inspection of in-service aircraft. To achieve the proof-of-concept of the proposed Smart Hangar, the advanced UPI system was developed with 20 kHz laser ultrasonic scanning system and real-time damage evaluation result generation to minimize aircraft inspection downtime. The 20 kHz laser ultrasonic scanning system consisted of a 532 nm Q-switched continuous wave laser (QL) and a laser mirror scanner (LMS) for ultrasound generation on a target. During the laser scanning, the LMS maneuvered the beam impinging point rapidly within a two-dimensional scan field at the maximum scanning frequency of 20 kHz. The propagating ultrasonic signal was interrogated by an ultrasonic sensing system, which consisted of an inline filter and amplifier, through either a noncontact, wired, or wireless ultrasonic sensor. In this paper, a wired ultrasonic sensor was used. Then, the ultrasonic signal was digitized in a DAQ system. Lastly, the digitized data was processed using multiple damage visualization algorithms and the damage can be analysed visually on a computer. The advanced UPI system was tested by inspecting two different carbon fiber-reinforced polymer (CFRP) structures at 20 kHz scanning rate. First, a CFRP wingbox with BVID was inspected with the scanning area of 520 mm × 520 mm. Second, a 1.36 mm thick CFRP plate with prepeg-gap due to debonding damage was inspected with the scanning area of 200 mm × 200 mm. Figure 2 shows that the advanced UPI system was able to detect and evaluate the damages. The quantitative results have obtained and showed good similarity to portable C-scan results. The results have shown the feasibility to implement the advanced UPI system to realize the Smart Hangar for large-scale inspection applications at high scanning rate.

1 INTRODUCTION

The usage of composite materials in aircraft structures has been increased. However, with increasing usage of composite materials in aircraft structures, damage cases of the composite structures also are occurring such as barely visible impact damage (BVID), delamination, debonding, and fiber-matrix breakage. To detect these damages and defects of composite materials, the visual inspection, the tap test, the ultrasonic testing and the thermography are generally used as non-destructive inspection (NDI) techniques, they also have many disadvantages. The ultrasonic propagation imaging (UPI) developed by J. R. Lee et al. is a damage visualization technique using a non-contact laser scanning system and ultrasonic sensor for structures such as aircraft, wind turbine

and others [1, 2]. This technique shows image results to arrange the data so that one spreadsheet corresponds to one image snapshot in the movie result, while one data point on each spreadsheet corresponds to one pixel of the image frame after acquiring ultrasonic signal data by an ultrasonic sensor to generate from a laser impinging point [1]. The UPI technique has many advantages such as rapid inspection speed, non-contact scanning, quantitative damage localization and evaluation. Furthermore, the long distance inspection at a few tens of meters is possible to collimate laser beam using a beam expander/collimator, it allows inspection of in-service large scale composite structures which is difficult to access by inspectors [2]. In this work, to achieve the proof-of-concept of the Smart Hangar, we developed the advanced UPI system with 20 kHz laser ultrasonic scanning system and real-time damage evaluation result generation to minimize aircraft inspection downtime.

2 DEVELOPMENT OF SMART HANGAR SYSTEM

Figure 1 shows the Smart Hangar, which is incorporated with an advanced ultrasonic propagation imaging (UPI) system to facilitate the full-scale inspection of in-service aircraft.

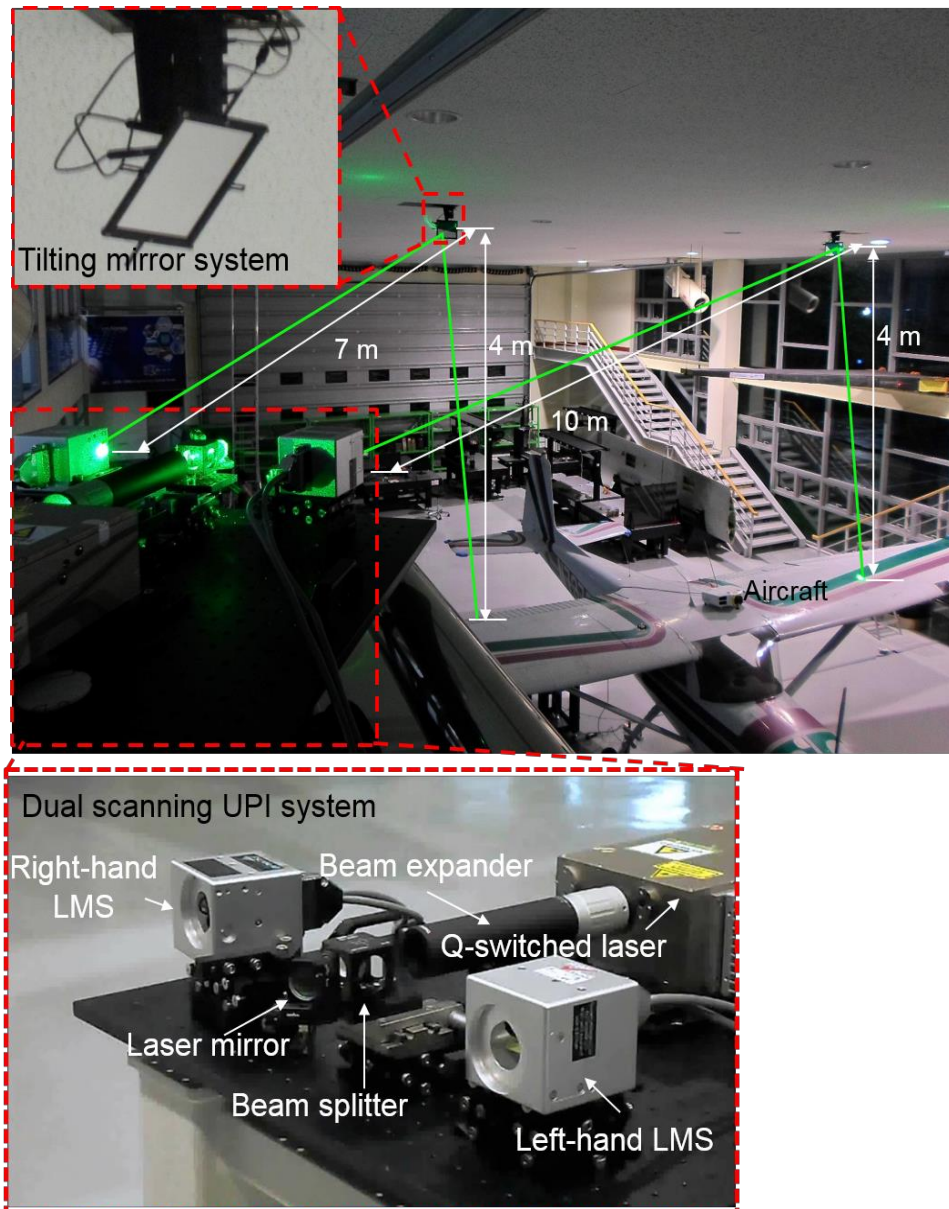


Figure 1: A technological demonstration setup of Smart Hangar.

As shown in Figure 2, the 20-kHz laser ultrasonic scanning system was used to generate laser beam using 532 nm Q-switched continuous wave laser (QL) and a laser mirror scanner (LMS). High speed scanning was performed on the scan area of the inspected specimen with a maximum scanning frequency of 20 KHz [3]. The ultrasonic signals generated by the laser impinging were examined using an ultrasonic sensing system, which consist of an inline filter and amplifier, through either a noncontact, wired, or wireless ultrasonic sensor. In this paper, a wired ultrasonic sensor installed on the opposite side of the target surface under the concept of embedded structural sensor was utilized. These ultrasonic signals were digitized in a high speed data acquisition (DAQ) system integrated with the processing unit. The digitized data was finally processed using a number of damage visualization techniques to analyse the impairment on the inspected area.

3 EXPERIMENTAL RESULTS

The advanced UPI system was tested by inspecting two different carbon fiber-reinforced polymer (CFRP) structures at 20 kHz scanning rate. Experimental setup of the advanced UPI system is shown in figure 3.

First, a CFRP wing specimen with BVID (Barely Visible Impact Damage) was inspected with the scanning area of 520 mm × 520 mm, scanning interval of 1 mm and a stand-off distance is 5 m as a shown in figure 4(a). The amp-integrated PZT sensor was installed on the reverse side of wing skin in the scan area. As shown in figure 5 (a), an ultrasonic wave propagation imaging (UWPI) result shows incident wave with scattering wave induced by a BVID and flange legions. However, this imaging method is difficult to evaluate damage size. To evaluate the damage quantitatively, Ultrasonic wavenumber imaging technique were applied [4]. As shown in figures 10 (b), UWI results show quantitative damage size of 24 mm × 24 mm and shape of the damage. To improve reliability, we compared the UPI results with a C-scan image measured by the manual C-scan system using pulse-echo mode at 20 MHz as a shown figure 4 (b). The result show that the damage size of UWI result is exactly similar to damage size of the C-scan result.

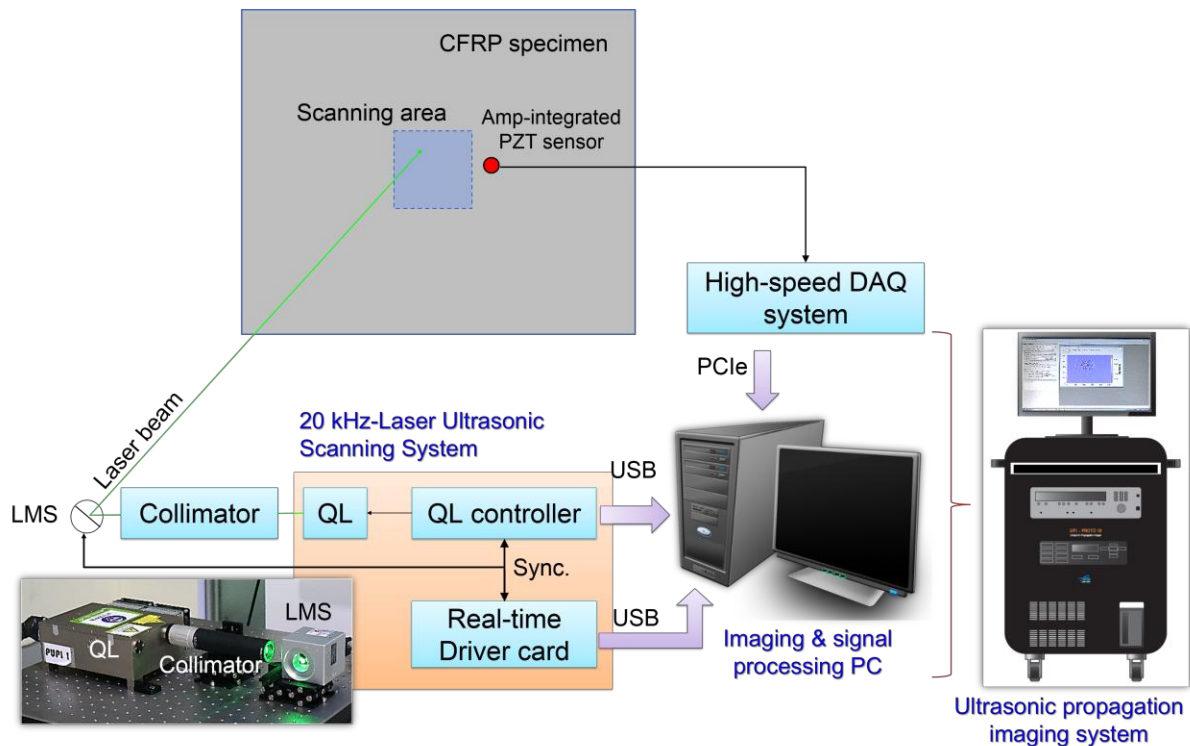


Figure 2: Configuration of the advanced UPI system with 20-kHz laser ultrasonic scanning system.

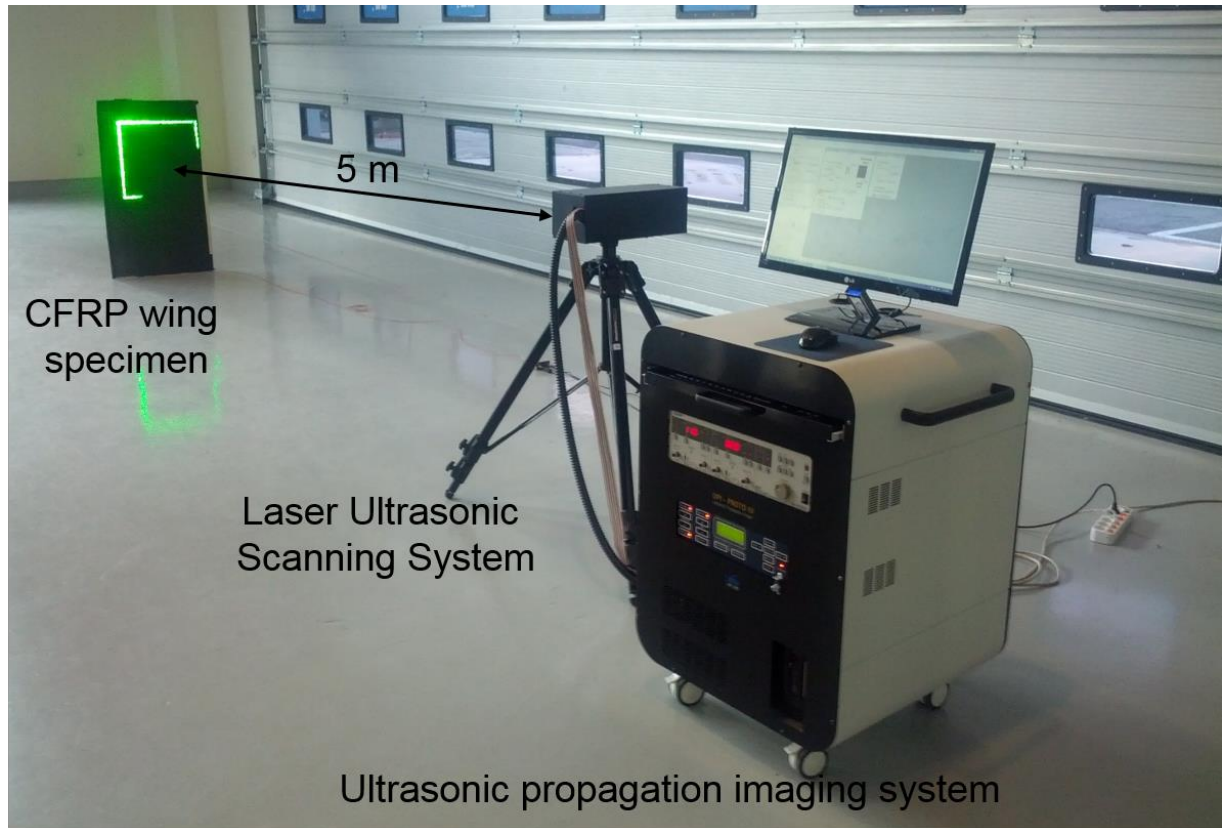
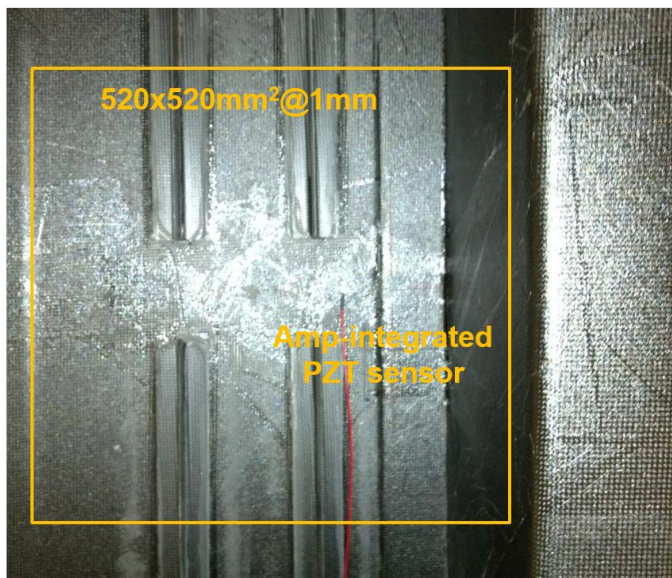


Figure 3: Experimental setup of the advanced UPI system with 20-kHz laser ultrasonic scanning system.

(a)



(b)

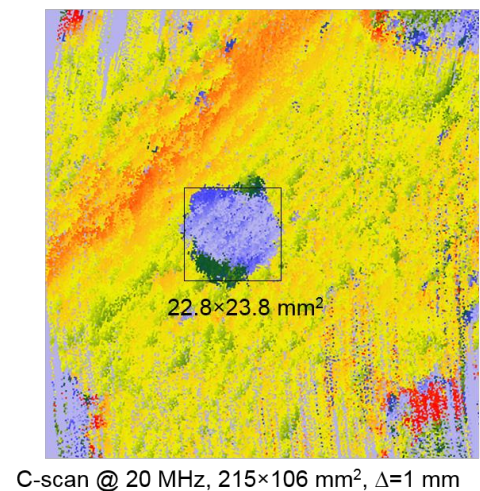


Figure 4: (a) CFRP wing specimen, (b) C-scan result of CFRP wing specimen.

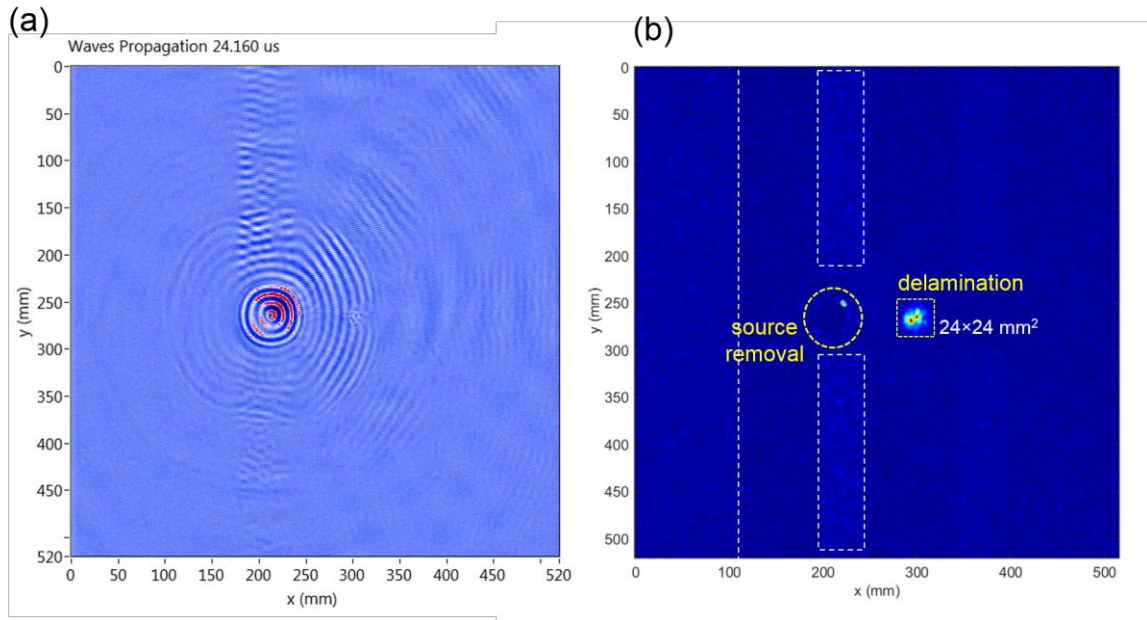


Figure 5: (a) UWPI (Ultrasonic wave propagation imaging) result, (b) UWI (Ultrasonic wavenumber imaging) result.

Second, a 1.36-mm thick CFRP plate with an internal prepreg-gap was inspected with a scanning area of $200\text{ mm} \times 200\text{ mm}$, scan interval of 0.5 mm and a stand-off distance of 5 m as shown in figure 6(a). The sensor was installed at the centre line of the scan height and 50 mm side the right-side limit of the scan area.

Figures 7(a) shows the ultrasonic wave propagation image at $134\text{ }\mu\text{s}$. The boundary of the prepreg gap line is showed by incident wave with scattering wave evidently. Variable time window amplitude mapping (VTWAM) was applied to evaluate the damage size [5]. As shown in figure 10 (b), VTWAM results show quantitative prepreg gap length of 70 mm. The C-scan result is also shown in Figure 6(b) where the damage length is estimated at 70 mm. The VTWAM result showed good similarity to the portable C-scan result. These results have proved the feasibility of the UPI system to realize the Smart Hangar for real world applications.

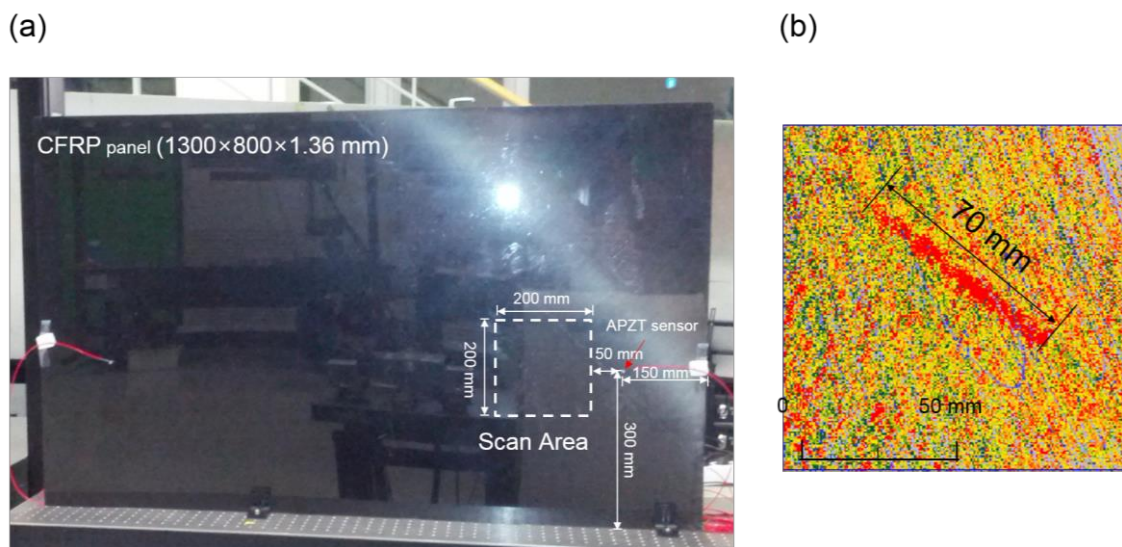


Figure 6: (a) CFRP panel specimen, (b) C-scan result of CFRP panel specimen.

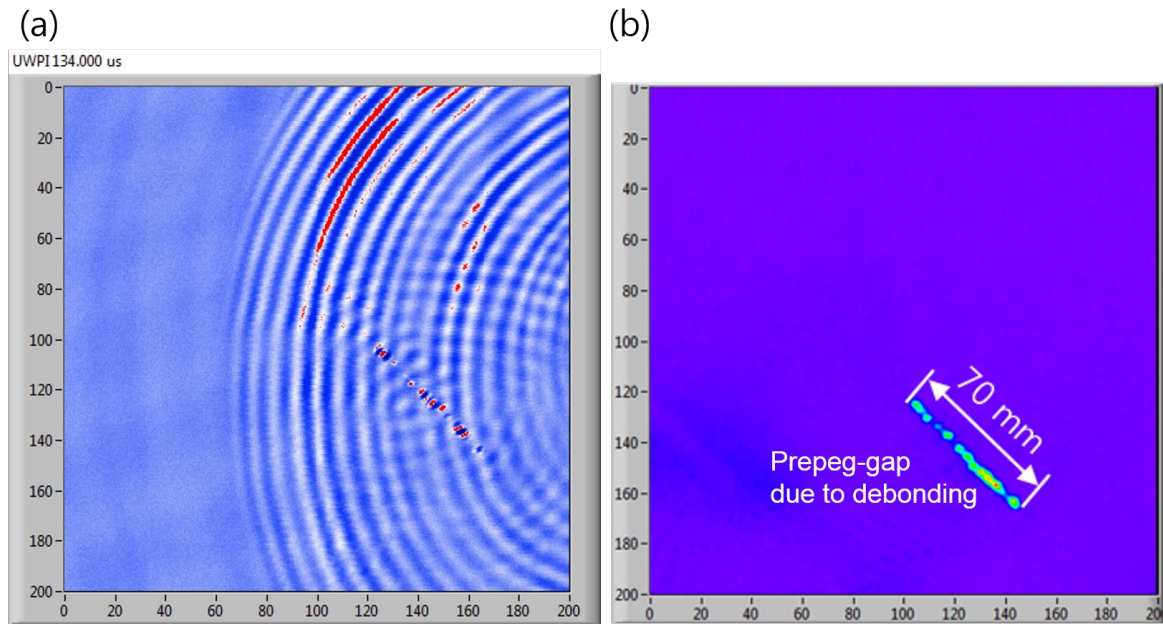


Figure 7: (a) UWPI (Ultrasonic wave propagation imaging result, (b) VTWAM (Variable time window amplitude map) result.

4 CONCLUSION

In this work, to achieve the proof-of-concept of the proposed Smart Hangar, the advanced UPI system was developed using 20 kHz laser ultrasonic scanning system and high-speed DAQ and signal processing system. First, a CFRP wingbox with BVID was inspected with the scanning area of 520 mm $\times$ 520 mm, UWI results show quantitative damage size of 24 mm $\times$ 24 mm and shape of the damage. To improve reliability, we compared the UPI results with a C-scan image, the result show that the damage size of UWI result is exactly similar to damage size of the C-scan result. A 1.36 mm thick CFRP plate with prepeg-gap due to debonding damage also was inspected with the scanning area of 200 mm $\times$ 200 mm. As a results, damage visualization results show quantitative prepeg gap length of 70 mm and showed good similarity to portable C-scan results. The results have shown the feasibility to implement the advanced UPI system to realize the Smart Hangar for large-scale inspection applications at high scanning rate.

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